Dylan Burkey

(404) 358-3445 | dburkey3@gatech.edu | linkedin.com/in/dylaneburkey | dylanburkey.netlify.io

Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Mechanical Engineering (Concentration in Manufacturing) May 2027
GPA: 3.24 (cumulative) and 3.54 (major)

Experience

Amazon May 2026 – Aug. 2026
Amazon Robotics Area Manager *Miami, FL*

- Collaborated with Reliability and Maintenance Engineering teams to commission new sensor hardware on CARDINAL's conveyor retro-reflective photo eye system; improved induct automation by increasing pallet stacking throughput by 8% and tote FPY by 15%.
- Led cross-functional NPI of a freight management application (Python/JavaScript) to bridge operational visibility into late packages; embedded automated task routing and audits into standard workflow to reduce freight rollover by 11.3k packages/month; implemented in 5 NA FC's.
- Coordinated 100+ associates across engineering and operations in a large-scale industrial facility; documented process improvements.

Smurfit Westrock May 2025 – Aug. 2025
Manufacturing Engineering Intern *Mooresville, NC*

- Designed and executed test plans across die cutter production runs; analyzed anvil cover wear patterns against die geometry using statistical methods; implemented a data-driven replacement schedule reducing downtime by 25% and extending blade service life by 10%.
- Created and validated engineering drawings and CAD models against ASME design standards for manufacturability; partnered with the mechanical design team using SOLIDWORKS and ArtoisCAD to verify new packing products for production on high-speed converting equipment.
- Created an inventory management application to track stock levels, analyze consumption patterns, and flag excess inventory for disposition, reducing stockout-related production downtime by 30% to support maintenance teams.

Snapping Shoals EMC Aug. 2022 – May 2023
Automotive Engineering Intern *Covington, GA*

- Performed mechanical diagnostic/repairs across a fleet of 75+ utility vehicles; managed inventory and procurement to minimize downtime.

Projects

Toothless: Autonomous Robot Aug. 2025 – Dec. 2025

- Designed and fabricated an autonomous mobile robot integrating machined, 3D-printed, laser-cut, pneumatic, actuating and sensor components with Arduino-based mechatronic control; placed 2nd out of 70+ teams in Georgia Tech's Creative Decisions & Design competition.
- Applied DFM/DFA and DFMEA through iterative prototyping of components with GD&T specifications in SOLIDWORKS/OnShape.

Case Western Reserve Univeristy - Bearing Vibration Analysis Jan. 2026 – Mar. 2025

- Diagnosed bearing faults on a 2-hp motor vibration data in MATLAB (RMS, regression, Newton-Raphson); estimated speed within 0.3%.
- Predicted bearing remaining useful life via numerical integration (RK4, Euler); validated within 0.1 hrs of true failure time.

Pioneer DDJ-FLX10 Rekordbox LCD Display Feb. 2026 – Present

- Created a custom touchscreen display module for the DDJ-FLX10 to provide an ergonomic Rekordbox interface and seamless navigation.
- Designed and 3D printed a custom enclosure and controller mount in SOLIDWORKS to house a Raspberry Pi LCD and display electronics.

Involvement and Leadership

Yellow Jacket Aviation Feb. 2023 – Present
Manufacturing Lead

- Created assembly plans and BOMs for manufacturability of an RV-14 aircraft build; mapped subsystem dependencies to eliminate rework risk.
- Coordinated across engineering, procurement, finance, and operations teams to align build scope with FAA regulations and yearly budget.

Curious Cardinals Dec. 2023 – Present
Technical Mentor

- Mentored aspiring mechanical engineering students in creating CAD and robotics project portfolios to support professional development.
- Developed a 10-week guide to teach STEM fundamentals and college readiness, currently serving 200+ active students in the country.

American Society of Mechanical Engineers Sep. 2023 – Present
General Body Member

- Participate in weekly professional development events and networking sessions with engineering companies to enhance industry knowledge.

Kappa Sigma Fraternity Aug. 2024 – May 2026
Social Chair

- Administered a $40K semesterly budget, managing vendor negotiations, cash flow tracking, and logistics for 50+ social events for students.

Technical Skills

Software: SOLIDWORKS, Onshape, Autodesk Fusion, AutoCAD, Ansys (Mechanical & Granta), Arduino, MATLAB, Minitab, Microsft Office
Hands-On: Mill and Lathe, Bandsaw, Laser Cutting, 3D Printing, Power Tools, Precision Instruments
Fundamentals: Physics/Mechanics, FEA, Thermal/Fluid/Heat Sciences, Design, Materials Science, Statistics, Circuitry, Economics
Operational Excellence: Lean Six Sigma, DMAIC, Lean Manufacturing, Project Management, Continuous Improvement, Kaizen, RCA
Certifications: Lean Six Sigma Yellow Belt, Amazon AI and Machine Learning Fundamentals, CompTIA A+, FE/EIT Expected 2027